

Revision Notes for Class 7 Science Chapter 10 Life Processes in Plants

All living things need food to grow, develop, and get energy for their life processes. While animals get their food by eating other living things, plants make their own food using simple substances from nature. Plants also grow in size, develop new leaves, and change their structure during their life span, showing clear signs of growth similar to animals.

How Do Plants Grow and What Do They Need?

Plants require certain essential factors to grow, mainly sunlight and water. Simple activities show that a plant kept in sunlight with regular watering grows best, while one without water wilts even if it gets sunlight, and a plant kept in darkness with water stays unhealthy. So, both sunlight and water are important for healthy plant growth.

How Do Plants Get Their Food?

Unlike animals, plants do not eat their food. Instead, green leaves make food for the entire plant. They use components like carbon dioxide from air, water from soil, and energy from sunlight to prepare a kind of sugar called glucose, which gets stored as starch.

The leaves have a green pigment called chlorophyll, which traps sunlight to help prepare food. This is why leaves are sometimes called the “food factories” of plants.

Role of Sunlight, Chlorophyll, Water, and Air in Photosynthesis

Several experiments show that leaves kept in the dark do not make starch, but those exposed to sunlight do. Only the green areas of leaves (which have chlorophyll) turn blue-black in a starch test after adding iodine, proving that chlorophyll and sunlight are both necessary for making food.

Another important part is carbon dioxide from the air. When a part of a leaf is kept in a closed bottle without carbon dioxide, that region cannot make starch. Only the part exposed to air (with carbon dioxide) makes food. Therefore, for photosynthesis, plants need four things: sunlight, water, carbon dioxide, and chlorophyll.

Plants take in carbon dioxide through tiny pores on their leaves called stomata. These also help with the release of extra oxygen, which is a byproduct of photosynthesis.

Process and Word Equation for Photosynthesis

In the process called photosynthesis, plants use sunlight to convert carbon dioxide and water into glucose (a simple sugar) and oxygen. The general word equation is:

Carbon dioxide + Water (in the presence of sunlight and chlorophyll) → Glucose + Oxygen

This process mainly happens in the leaves, but other green parts of plants can also perform photosynthesis. The oxygen released helps almost all living things survive and keeps the air balanced.

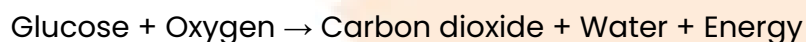
Transport of Water, Minerals, and Food in Plants

Roots absorb water and dissolved minerals from the soil and transport them up to the rest of the plant through special tubes called xylem. This water is not just used in food preparation but also for other functions like cooling and nutrient transport.

Once leaves produce food, it must reach other parts of the plant such as roots, stems, and fruits. Food is transported by another set of tubes called phloem. The movement of water and food throughout the plant is essential for its survival.

Respiration in Plants

Just like animals, plants also breathe, or respire. They use up the glucose made during photosynthesis and oxygen from the air to release energy. This process also gives off carbon dioxide and water. The word equation for plant respiration is:



This energy is used by plants to carry out their life processes such as growth, repair, and movement of materials. Respiration happens both day and night, while photosynthesis only takes place in the presence of light.

Photosynthesis vs Respiration

Feature	Photosynthesis	Respiration
Raw materials	Carbon dioxide, water, sunlight, chlorophyll	Glucose, oxygen
Products	Glucose, oxygen	Carbon dioxide, water, energy
Word equation	$\text{CO}_2 + \text{H}_2\text{O}$ (light, chlorophyll) $\rightarrow$ Glucose + O_2	Glucose + $\text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O} + \text{Energy}$
Importance	Provides food and oxygen	Releases energy for life processes

Interesting Facts and Activities

Historical texts like “Vrikshayurveda” from ancient India have described plant growth and farming methods centuries ago. Modern scientists from India, including Rustom Hormusji Dastur and Kamala Sohonie, made significant contributions to understanding photosynthesis and plant nutrition.

Through simple activities—such as testing for starch in leaves, checking for oxygen bubbles in water plants under sunlight, and comparing plant growth in different conditions—students can observe various plant processes directly at home or in class.

How Do Leaves Exchange Gases?

Leaves have tiny pores called stomata, mainly on their lower surface. These pores allow carbon dioxide to enter for photosynthesis and oxygen to move out. They also help plants “breathe” for respiration. Observing leaf peels under a microscope shows these pores clearly.

In Brief

- Plants make their own food using sunlight, water, carbon dioxide, and chlorophyll in leaves.
- This process, called photosynthesis, also releases oxygen.
- Xylem carries water and minerals from roots, phloem carries food from leaves to the rest of the plant.
- Stomata in leaves allow exchange of gases.
- Plants also “breathe” by breaking down food to get energy, which is used for their growth and development.